

From machinery to sensibility: (Un)learning cyanobacteria between biological laboratory practice and more-than-human exploration

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Pictorial

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Abstract

Biodesigners are increasingly recognising the limitations of purely utilitarian approaches to designing with living organisms when pursuing the broader, holistic regenerative aims of biodesign. Yet developing more inclusive, self-reflective practices that consider the well-being of other beings remains challenging, as it requires unlearning deeply embedded anthropocentric assumptions and modes of thinking. In response, we present a three-year self-reflective account of a biodesigner navigating between protocol-driven laboratory experimentation and an exploratory more-than-human design approach in a studio-lab hybrid setting while working with cyanobacteria. We employed augmented and immersive digital tools to support the biodesigner's journey—not only to cultivate new sensibilities, but also to enable shifts in scale and perspective beyond the constraints of laboratory experimentation. Through rich visuals and storytelling, and drawing on an autoethnographic approach, we compare these modes of practice to articulate microbially attuned perspectives and multispecies-aware approaches, revealing both opportunities and challenges for integrating more-than-human sensibilities into biodesign.

Prologue

Since the publication of Myers' foundational work in 2012, which positioned biodesign as a novel design paradigm integrating living organisms to generate novel functions, forms, and sustainable modes of production, the field has expanded rapidly. This expansion has been driven by technological advances and the acceleration of interdisciplinary collaboration (Atzmon 2023; Karana 2020; Marseglia *et al.* 2025). Today, biodesigners increasingly realise, upscale, and disseminate their ideas through education and public engagement, broadening the field's scope (Grushkin *et al.* 2022; Toivonen and Van Ditmar 2024; Walker *et al.* 2023). Alongside these developments, ethical awareness and critical discourse surrounding biodesign have evolved in parallel, including calls for more holistic and regenerative approaches (Armstrong 2022; Collet 2020; Karana *et al.* 2023). These perspectives advocate fostering creative assemblages; environments where humans and non-humans actively participate and coevolve within a dynamic ecology of interconnected living and non-living entities (Karana *et al.* 2023).

As biodesign is applied more substantively in real-world contexts, these considerations have shifted from foresight concerns to practical and urgent questions for the near-term development of the field. Biodesigners are increasingly called upon to engage with the full lifecycle of living artefacts, while attuning to more-than-human (MTH) temporalities and scales (Ikeya *et al.* 2025; Ofer and Alistar 2023; Zhou *et al.* 2024), and multispecies interactions that unfold across these stages (Groutars *et al.* 2024). Yet most biodesign practices, particularly at the microbial scale and within applied research contexts, remain dominated by heavily controlled scientific workflows (Ihls and Pollini 2025). While these approaches ensure scientific rigour, technical feasibility, and quality control, they often entail *implicit instrumentalisation* (Zhou 2025). Even with notable advancements in MTH design methods (Giaccardi *et al.* 2024; Nicenboim *et al.* 2024), their implementation in biodesign laboratory settings remains limited. **To address this gap, we engage directly with biodesign practice to explore whether and how MTH sensibilities can be cultivated and enacted alongside conventional laboratory conditions.**

The first author (biodesigner) approached this inquiry as a designer with an artistic background, newly engaging with laboratory-based biological research. Prior to the three-year engagement described here, their experience with biology and microbial systems was limited, having developed an interest in ecological and microalgal sustainability through design but only beginning to learn biological fundamentals through a graduate biodesign course at Delft University of Technology. Immediately following this foundational exposure, the first author entered the biodesign lab to explore scientific protocols and cultivate hands-on familiarity with living systems. The lab trajectory commenced at this point, focusing on in-depth exploration of a specific cyanobacterial strain—*Synechocystis* sp. PCC 6803 (Stanier *et al.* 1971)—which formed the basis

for subsequent design inquiry. Over a period of three years, the account traces a process shaped by both in-situ, contemporaneous recording and note-taking, and retrospective synthesis. Experimental milestones, laboratory notebooks, logs, photographs, and ongoing personal and dialogic reflections collectively scaffolded the inquiry, weaving together immediate impressions and reflective insights. Its biological traits, particularly phototactic behaviour, prompted experiments focused on motility and scale. These investigations developed along two parallel trajectories: one, the macroscopic manifestations of motility, guided by a biological protocol, and two, the speculation on light sensing in relation to movement through an experiential mode of *becoming-with* cyanobacteria (Kim *et al.* 2024). The speculative trajectory emerged after a period of laboratory immersion, as the first author began to work more specifically with cyanobacteria and grappled with the question of how these organisms might sense the environment around them. While Kim *et al.* (2024) laid the conceptual groundwork for cultivating MTH sensibilities toward microbes in biodesign, the present study introduces and demonstrates specific operational scenarios for enacting and examining these sensibilities in the laboratory context.

The first author's experiences are articulated through two autoethnographic accounts that foreground moments of self-reflective (un)learning within laboratory and studio–lab hybrid settings. These accounts make visible the methodological boundaries of contemporary biodesign practice and explore how biodesigners can embody MTH sensibilities from within existing regulated conditions. Many MTH works use decentring methods that immerse practitioners in shared environments and enable them to access knowledge often inaccessible to humans alone (Giaccardi *et al.* 2016; Özgen *et al.* 2025; Rossellini 2010; Thwaites 2016), and the first author similarly positioned themselves alongside microbes to gain a deeper, more acclimatised understanding within their experimental context. The first trajectory investigates cyanobacterial motility through laboratory studies supported by a bio-digital setup (phototaxis toolbox) for tuning light conditions, along with an optical see-through head-mounted display that was primarily used to access scientific literature during the lab experiments. The second trajectory involved a video see-through head-mounted display to visually speculate on cyanobacterial light response.

Through this journey, we critically examine how engaging with cyanobacteria involves both **learning established scientific protocols and unlearning anthropocentric assumptions embedded within them**. Early in the process, moments arose that unsettled the first author's prior approaches: for instance, repeated failures to induce cyanobacterial motility in the laboratory prompted a reevaluation of seemingly minor environmental factors, and illuminated the need to let go of human-centred expectations of control. Similarly, the attempt to simulate cyanobacterial sensing through immersive digital tools highlighted the limits of translating microbial sensory experience into human terms, underscoring the need to unlearn ingrained visual biases. These early experiences prompted a reconsideration of ingrained habits and exposed a broader methodological limitation: conventional laboratory documentation rarely makes space for the complex relational or contextual factors in biodesign. Together, they point to something more fundamental—practising with more-than-human sensibilities demands not just new methods, but a shift in perspective that requires letting go of control, unlearning established conventions, and cultivating new modes of attention, articulation, and recognition.

Negotiating between instrumental and MTH perspectives

What follows are two parallel trajectories of a biodesign research journey, presented side by side across these pages. On the left, the laboratory trajectory unfolds through biological experiments following established research protocols. On the right, the design studio trajectory moves outside the lab, where speculative explorations take centre stage. While each begins from different starting points, both ultimately encounter moments of (un)learning as their trajectories unfold. Both are narrated from the autoethnographic perspective of the first author. Autoethnography is an ethnography focused on the self where the author is both informant and investigator (Cunningham and Jones 2005). Also known as first-person research, it enables designers to create for themselves and refine their work through personal use (Höök *et al.* 2018; Lucero *et al.* 2019; Neustaedter and Sengers 2012). Given that it has proven to be useful for capturing everyday nuances and is well-suited to design's hands-on, embodied nature (Adams *et al.* 2013), while encouraging critical reflection (Desjardins and Wakkary 2016; Tomico *et al.* 2023), it is an excellent methodology for this inquiry. In biodesign, autoethnographic approaches have been used to foreground first-hand laboratory experiences, revealing the interdisciplinary dynamics and the drifting boundaries of knowledge in this field (Kuznetsov *et al.* 2018; Sawa 2016; Stefanova 2021). Past efforts have shown the value of integrating laboratory observations and lived experience, merging personal standpoints with empirical insights (Pollini 2024). Building on this foundation, we use an autoethnographic lens to probe actionable MTH modes of practice within the biodesign lab operating under regulated scientific backdrops. At its core, our autoethnography is a self-reflective journey marked by uncertainty, vulnerability, and permeability. Our accounts intentionally foreground these qualities, adopting a direct and subjective voice to reflect the lived experience of inquiry. The tone and subheadings are intentionally unpolished, capturing the first author's ongoing tensions and negotiations between instrumental aims and MTH perspectives throughout the process. Visual cues throughout the narrative are used to highlight and trace the first author's nuanced moments of  tension,  negotiation,  unlearning,  perspective shifts, and  human-centred assumptions.

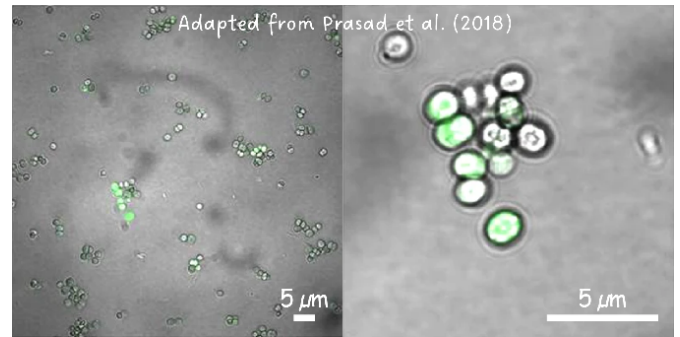
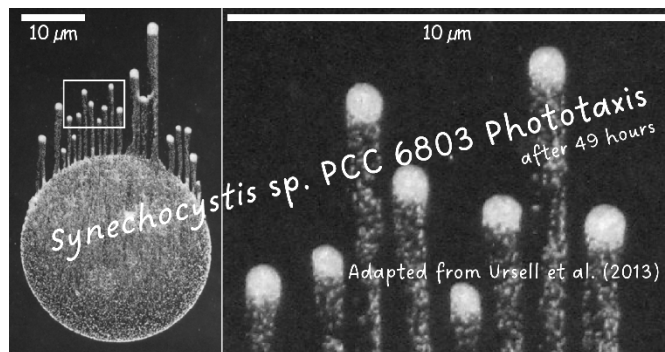


So, what is *Synechocystis* sp. PCC 6803?

As someone trained in design and learning about biology, my encounter with *Synechocystis* sp. PCC 6803 was not intentional. PCC 6803 is a unicellular freshwater cyanobacterium capable of phototrophic growth via oxygenic photosynthesis in light and heterotrophic growth in darkness (Heidom et al. 2011). Recommended by the biodesign lab technician (Joren Wierenga) and the fourth author, a cyanobacteria specialist, my initial reaction was negative: memories of toxic algal blooms, the acrid smell of out-of-control eutrophic water, and green slime coating surfaces in my fish tank. Yet, seeing these organisms under controlled laboratory conditions changed my perspective. Instead of foul odours, I found vivid green liquid, quietly supporting the biodesign work at hand. Cyanobacteria are ancient world-makers, having existed for approximately 2.7 billion years and contributing to the Great Oxygenation Event (Demoulin et al. 2024). Beginning this research became my first positive interaction with cyanobacteria. In my lab explorations, I learned to culture PCC 6803 in BG11 medium under cool-white light at $\approx 20 \mu\text{mol photons m}^{-2} \text{ s}^{-1}$ and at room temperature (20–22°C) (Rippka et al. 1979; Rippka 1988; Williams 1988), observing the gradual intensification of the green in the culture flasks as their photosynthetic pigments accumulated. All these qualities made it an intriguing, if humbling, entry point for my laboratory journey.

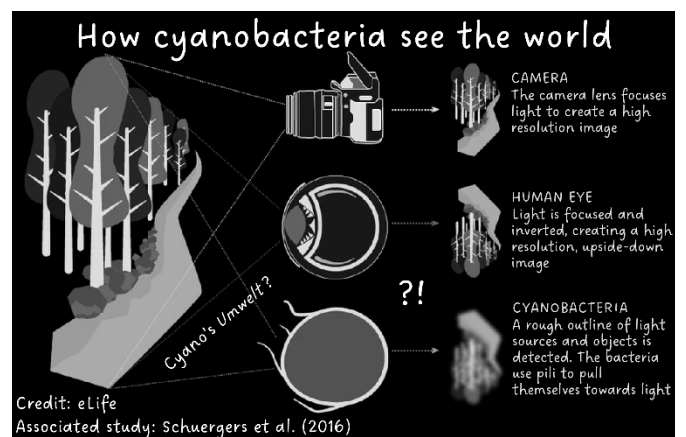
What, they can move?!

I had no idea they could move. The discovery arrived unexpectedly while exploring basic biological practices with PCC 6803; I encountered a phototaxis assay conducted at both macroscopic and microscopic scales (Jakob et al. 2017). The images showed green streaks spreading across agar surfaces, microscopic living beings moving in ways visible even to the human naked eye. The moment felt improbable, marked by a temporal dissonance between microbial movement and human perception, motion unfolding at a scale I was not expected to notice. The technical explanations came later, grounding my wonder in this movement. They use retractile type IV pili (T4P) to traverse moist surfaces in a jerky motion called twitching motility (Bhaya et al. 2000), and individual cells can directly detect unidirectional illumination by focusing light in a sharp focal point on their distal side (Schuergers et al. 2016). But what captivated me was simpler: the possibility of observing their movement toward light at a macroscopic scale, like plants bending toward windows or insects drawn to bulbs. This visibility—from cellular machinery to behaviour potentially observable without complex instrumentation—collapsed the distance I had assumed between myself and microbial life. This encounter with phototaxis became the foundation for what would follow in my laboratory practice.



What, they can see?!

While studying their motility, I became captivated by the mechanism enabling it: micro-optics. Each *Synechocystis* cell functions as a microscopic eyeball. Directional illumination produces a sharply focused spot of light—resembling a photonic nanojet—at the cell periphery opposite the light source (Schuergers et al. 2016). This lensing effect creates an intensity gradient that is much stronger than simple shading, allowing the cell to perceive light direction and control its movement accurately. At 3 μm in diameter—half a billion times smaller than the human eye—these cells perceive inverted, low-resolution images, discerning only blurred outlines of objects (Schuergers et al. 2016). The comparison, especially with a commentary illustration¹, inspired my design imagination and prompted me to verify its scientific validity with the study's authors and publisher. They “see”, not metaphorically, but optically. Their entire ‘cell’ operates as a sensory organ, collapsing the distinction between perceiving and being. This recognition shifted my experimental curiosity: if they possess ‘vision’—however alien to my own—could I technically inhabit their visual perspective? Could I design a way to see as similar as they seem to ‘see’, to experience light not as ambient illumination but as directional information guiding movement? The question moved me beyond observing their behaviour toward attempting something more intimate: becoming-with cyanobacteria through shared sensorial experience.

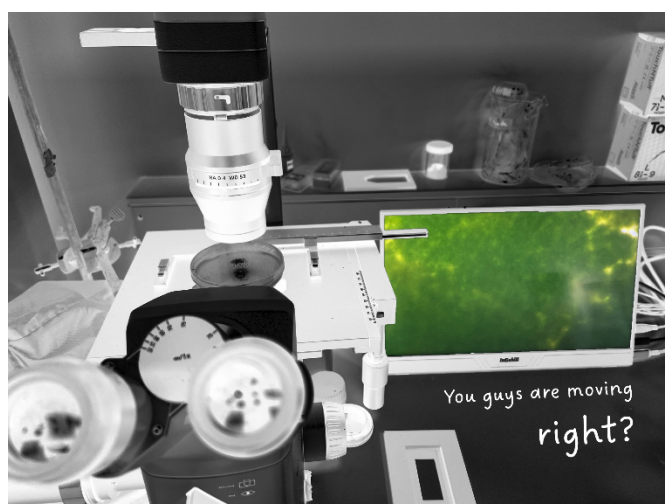


Okay, let' make them move

I was certain it would be simple to make PCC 6803 move. Full of excitement and with minimal protocol details, I began growing the cells in the lab to comprehend and test their phototaxis behaviour firsthand. I watched the culture flasks deepen from pale to vivid green, showing this organism's growth and readiness for experimentation. I prepared Petri dishes with their nutritional requirements (BG11 agar medium), folded black paper into makeshift light barriers, and inoculated the cultures I judged "green enough" onto the prepared surfaces. My understanding was reductive: healthy cyano, BG11 agar medium, directional light. I centrifuged cells to concentrate them into a deep, dark inoculum, confident that having more cells would help them to thrive and subsequently amplify their movement.

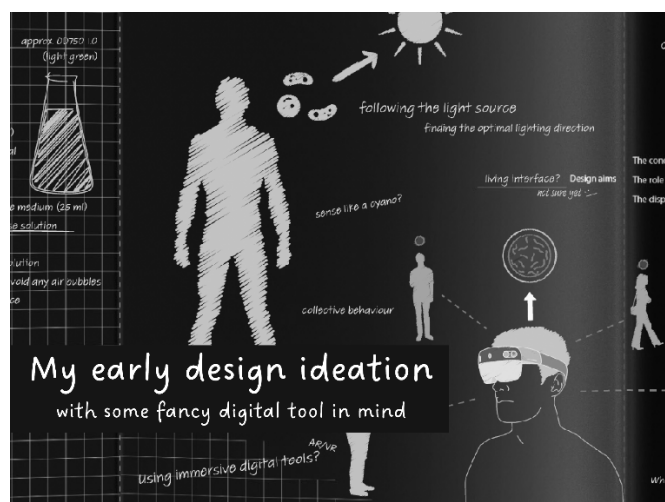


△ **They never moved.** Sometimes they bleached out, pale apparitions of their former green. Sometimes contamination bloomed, unknown fungi, heterotrophic bacteria claiming the plates I thought I'd prepared carefully. Sometimes they simply stayed green at the initial inoculation site, looking perfectly healthy under the microscope yet utterly immobile, as if refusing to meet my expectations. I had anticipated one month for this experiment. It took far longer than I anticipated. The failure was relentless, repetitive, humbling. Each unsuccessful plate became a lesson in what I had overlooked, dismissed, or misunderstood about the conditions these organisms required to move.



Okay, let's see like them

This question pulled me out of the laboratory and into design speculation. I began by depicting comparative visions: if they see through their entire cell as a lens, what would colour, resolution, focal length, and field of view mean from their perspective? The literature offered sparse visual information, mostly microscope-enhanced imagery and an illustrative figure (Schuerger et al. 2016), which I relied on heavily. I sketched possibilities: an apparatus that would let me perceive surroundings through cyanobacterial light sensitivity, experiencing illumination not as an ambient backdrop but as a directional imperative. Could I design a wearable system that translates their phototactic response into a human sensory experience? More provocatively: if they move collectively in response to light gradients, could I create a human assemblage—a nexus of people wearing synchronised devices—that behaves like a colony? The speculation felt simultaneously absurd and necessary.

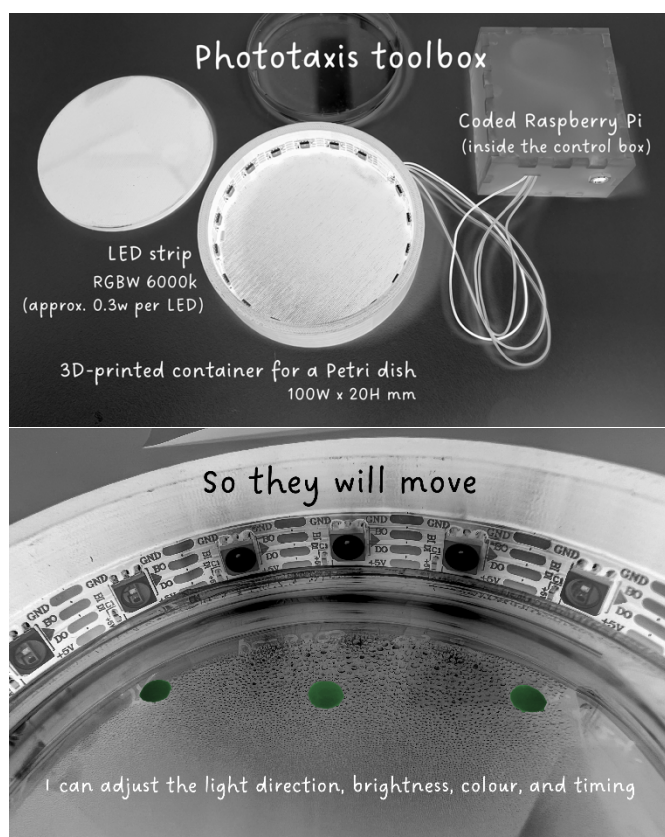


🌀 I was attempting to imagine MTH modes of knowing the organisms I worked with, not through observation but through embodied simulation. Immersive digital tools—specifically, video see-through head-mounted displays (HMDs)—offered technical means to construct perceptual experiences beyond human sensory norms. The question was no longer whether I could observe their vision, but whether I could inhabit it, however partially and imperfectly.



So I built an augmented setup

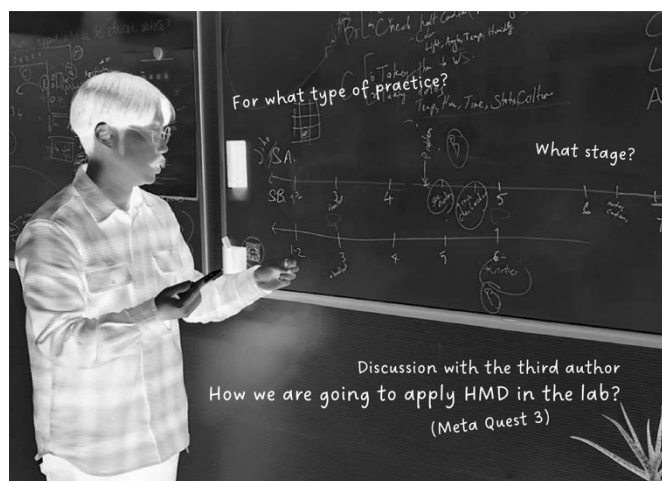
The failures forced me into conversation. I met repeatedly with the biodesign lab technician, who guided me through the lab experiments and with the fourth author, a microbiologist, cataloguing what I had overlooked: temperature, humidity, light intensity, sterile conditions, precise medium type, agar concentration, and even strain accuracy. Each variable I had dismissed as minor could derail everything. Meticulousness, I realised, was not pedantry but attentiveness, a form of respect for microbial requirements I had failed to extend. From that moment, I sought to understand the shared environment at their microbial scale, attuning my experimental setting to their level of sensibilities. I began augmenting conventional lab tools with digital instruments: a customised phototaxis setup we had built from 3D-printed containers, LED strips, and a Raspberry Pi; a photometer and an infrared thermometer; and a HoloLens 2 to visualise data spatially during practice.



I also learned that PCC 6803 has several known wild-type variants that can differ greatly in characteristics such as motility. Sadly, many studies used distinct substrains, often without specifying them (Schmelling and Bross 2024), leaving me confused. As the protocol I was referring to also used a particular substrain, I requested the exact motile substrain, "PCC-M," from the protocol's author. The organisms I sought carried their own lineage and history of care.

So I built an immersive setup

I lacked the technical skills to implement this vision, so I sought collaboration with the third author, a spatial computing researcher curious about HMD applications in biodesign. I brought them into the laboratory—showed them the incubators, sterile workflows, and cultures I tended—and explained what I was attempting. We discussed translating speculative vision into technical possibility and chose accessible tools: Meta Quest 3 for hardware and Unity for development. Although the Quest 3 primarily supports fully immersive virtual environments, its passthrough capability offered sufficient see-through quality to function as a controllable perceptual filter rather than complete immersion. This mattered. I wanted to remain situated in the biodesign workplace while perceiving them through cyanobacterial sensory parameters, not escape into virtual abstraction.



We aimed to create a cyano-vision simulation I could wear in the lab and beyond. The timeline forced compromises; we had to prioritise which visual parameters to simulate and which to defer. Yet through iterative co-ideation and their technical support, the concept gradually materialised into something experienceable: a wearable apparatus that would mediate my perception, allowing me to move through familiar spaces while sensing light as directional information rather than ambient illumination, which we called *Cyano Vision*.



Let's make their motility controllable!

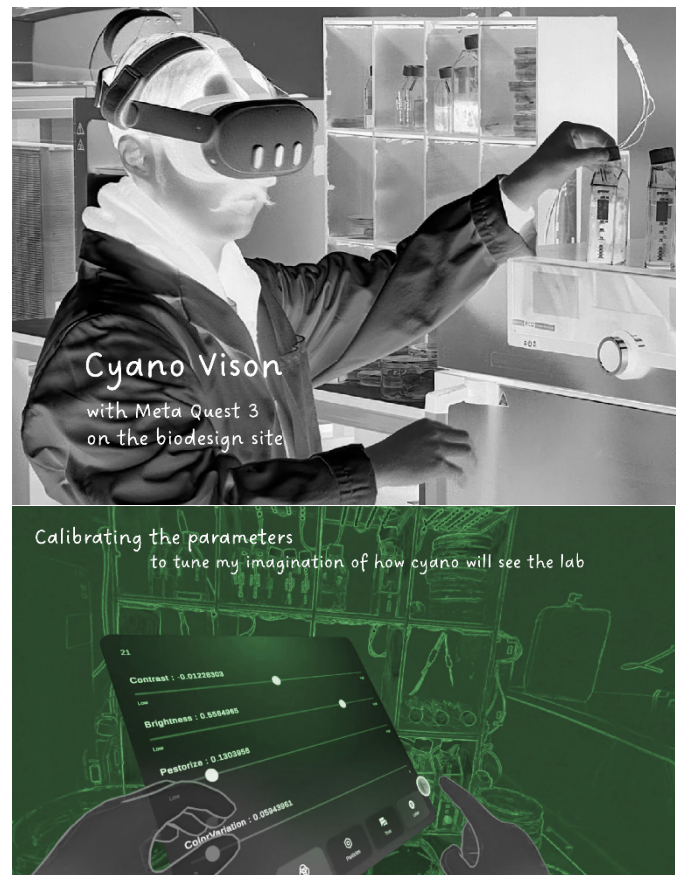
I felt ready. The failures had transformed me into someone who engaged with protocol as microbial requirements, not rules to follow, but sensitivities to respect. I had adapted the protocol to reflect both their needs and my available equipment. The augmented setup allowed me to attend to multiple scales simultaneously: automated light control, real-time environmental monitoring, and hands-free check of scientific details on existing literature while maintaining sterility.



🌀 **Everything felt aligned.** I was genuinely excited, anticipatory in a way that made the waiting feel charged with possibility. Would I finally see them move? My team shared this excitement. We gathered around the subtle promise of movement—not merely to witness motility under the lens, but to dwell on what such a quality might come to mean for biodesign. The conversations had already leapt ahead: if we could make them move in response to light cues, could we scale this up? Programme patterns? Create living displays? The instrumental possibilities felt tantalisingly close. I stood at the threshold between failure and hoped-for success, between observation and application. I carried all my preparation toward organisms I believed would finally perform for me. I was unaware that I was crossing from learning to see their autonomous behaviour toward seeking to make them move, to control their movement, to instrumentalise their capacities for human-determined ends.

Let's make their vision simulatable!

When I first wore the prototype of *Cyano Vision*, I felt astonished. After two years of speculation, the collaboration had produced something experienceable. The mixed reality simulator immersed me in a liquid-like environment: blurry, monochromatic blue-green vision, floating particles suspended in aquatic space. All parameters required manual adjustment; I could attune the filter toward microbial sensibilities (Kim et al. 2024), making it more alien, or clarify it for instrumental utility so I could continue laboratory work while wearing it. The simulation relied entirely on the Quest 3's passthrough cameras, making calibration subjective, guided by imagination rather than measurement. I was excited by what this gradual realisation enabled.



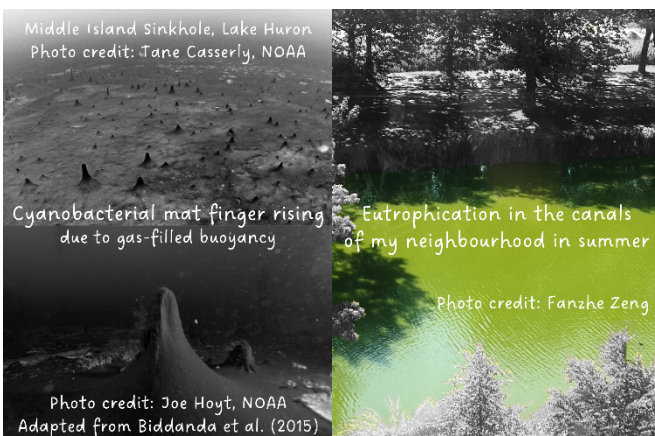
🌀 **Moving through the laboratory wearing the apparatus made the ontological distance between my sensory world and theirs palpable.** Each parameter adjustment forced me to articulate what I imagined their perception might be. I began proposing technical upgrades to make the simulation more scientifically accurate and immersive: sensors to measure light intensity, directional algorithms to respond to photon gradients, and features grounding the experience in empirical data. I imagined amplifying the somatic becoming experience, making the filter responsive to actual light conditions. The possibilities multiplied. If I could calibrate this apparatus accurately, it could become a tool for embodied understanding across different microbial scales and sensory modes.

Wait, does this happen in nature?

A question interrupted my instrumental momentum: Does this motility happen in nature? I began searching beyond the protocol. What I found exceeded my laboratory imagination. *Synechocystis* phototaxis in nature is questionable, since their lower light sensitivity means they respond only to bright sources like sunlight in shallow water, and microphysical factors further obscure their movement (Nilsson and Colley 2016). Yet, other phototactic cyanobacteria offered insights at scales I could witness. Some *Oscillatoria* species form massive conical mat structures through gas-filled buoyancy, and their collective phototactic movement can create visible architecture (Biddanda et al. 2015). In nature, cyanobacteria can grow to form visible green mats at the water's surface. This phenomenon, known as cyanobacterial blooms, is influenced by several factors, including light, which directs cells toward optimal photosynthetic conditions. As a result, biomass becomes concentrated, nutrient cycles shift, and water chemistry and colour are transformed (Dervaux et al. 2015; Wu et al. 2024; Xue et al. 2023).

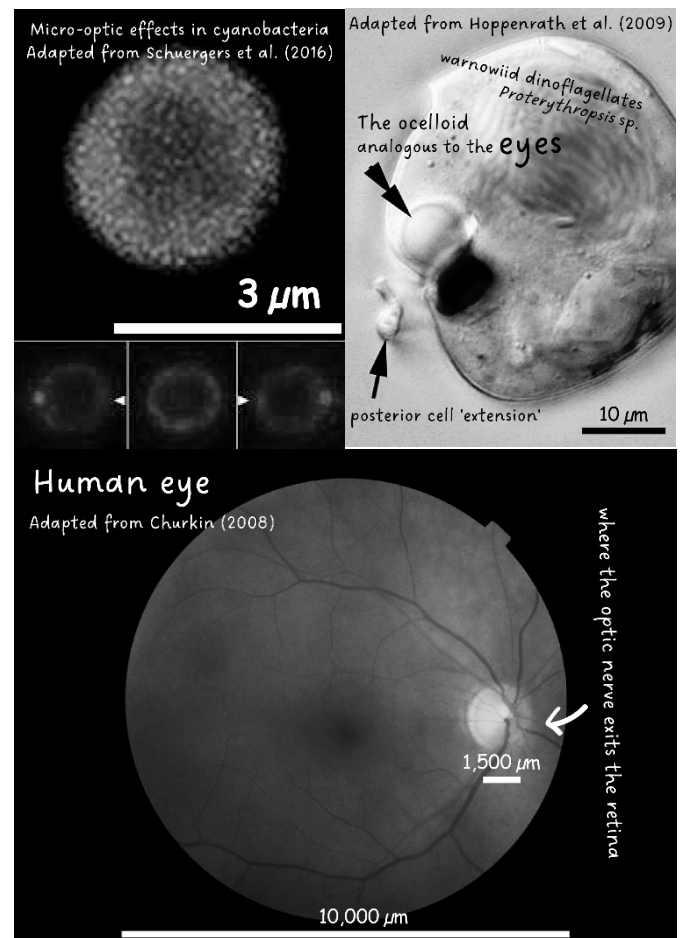


☞ What I had pathologised as environmental contamination revealed itself as ecological agency. Stepping outside the laboratory, I began to recognise their dynamic presence everywhere: seasonal cyanobacterial blooms coloured canals and lakes, even pushing people away from the water during summer. They were not performing for my apparatus; they were actively reshaping ecosystems, mediating planetary chemistry, operating as world-making agents whose movements affect and shift ecological processes across scales I was only beginning to comprehend.



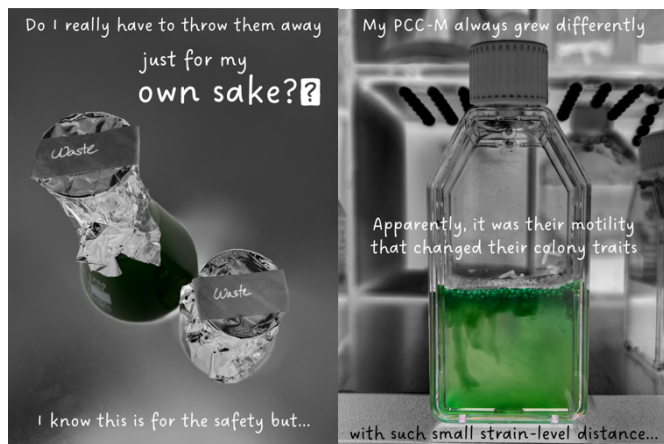
Wait, do they really “see” the world?

My collaborator (third author) asked a question that stopped me: ☞ “Why are you focusing on visualising their perception, when their understanding of seeing fundamentally differs from human vision? It’s actually about sensing light and direction, not vision in the human sense.” The question exposed my assumption. I returned to literature. The cell from *Synechocystis* does act as a lens and the surrounding membrane as retina, but this does not constitute a “real eye” in biological terms, because these structures have other essential functions that are not primarily devoted to vision (Nilsson and Colley 2016). In fact, the simplest organisms with real eyes are warnowiid dinoflagellates, unicellular eukaryotic algae measuring 50–100 μm , far larger than the 3 μm *Synechocystis* (Nilsson and Colley 2016). I had been translating their sensory world into human visual terms, privileging sight as the primary mode of knowing. But cyanobacteria don’t “see” in the ocular sense I had been simulating. They sense light. Their entire cell structure responds to directional photon flux as information guiding movement. The recognition troubled my entire approach. I had requested designing visual spectacle—immersive imagery, monochromatic filters, particle effects—when what I needed was sensory reorientation toward directional light response. The apparatus I had shaped mediated the wrong modality. My human-centredness shaped not only the simulation’s parameters but also its fundamental premise. In that moment, I realised that attuning to micro-scale conditions requires more than applying novel techniques; it demands ☞ unlearning deeply ingrained human ways of sensing.

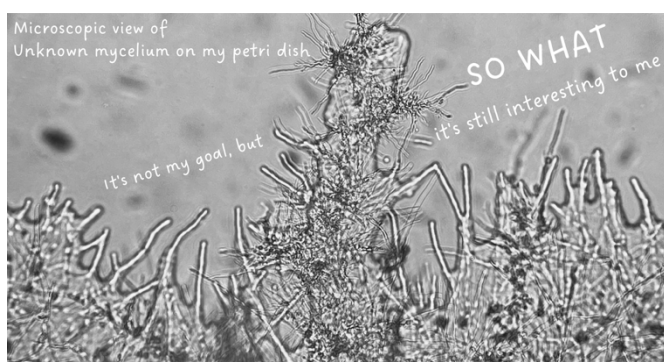


I still couldn't get them to move, but...

Returning to the laboratory, I confronted an irony: after nearly three years, I still could not make them move toward the light. Technical mastery remained elusive. Yet something more consequential had shifted. ☹️ **The repeated failures, the augmented instruments, and the oscillation between controlled laboratory conditions and in-situ ecological observation fundamentally reconfigured my relationship to these organisms.** Navigating the gap between strain and substrain taught me that even subtle genetic differences can yield distinct forms of livingness, highlighting the complexity and variability inherent to microbial life. Observing local wild cyanobacteria further emphasised that strain-level classification is a fragile, human-maintained construct, nearly impossible to pinpoint in nature. This tension unsettled the boundaries I had assumed between lab organism and ecological being, between experimental subject and autonomous world-maker. The friction between protocol-driven practice and field-based encounters surfaced ethical questions that biodesign often brackets: What does "contamination" mean from multispecies perspectives? How do practices of "waste" and "sterilisation," essential for biosafety and technical rigour, constrain the epistemic bearing and operational legitimacy of MTH sensibilities that regenerative futures demand?



☹️ I remain committed to technical mastery, but I now understand failure, augmentation, and scale-shifting not as obstacles to biodesign but as methodological conditions, ways of holding biodesign practices accountable to the microbial worlds they seek to engage.



I still couldn't see like them, but...

Returning to the design studio, I confronted what the prototype had revealed: ☹️ **I had designed for visual immersion when I needed sensory translation.** With my collaborator (third author), I began experimenting with light sensors and haptic feedback, assembling accessible components to capture light-related data through haptic feedback rather than visual-only sensing. We tinkered with different sensors and motors to improve our response when light conditions matched the biologically relevant range of *Synechocystis*, which is about 10 $\mu\text{mol photons m}^{-2} \text{s}^{-1}$ of directional illumination (Schuergers *et al.* 2016). The setup was almost becoming like a health monitor, but for cyanobacterial well-being rather than human metrics. Yet the immersive simulation had generated its own insights. The difficulty of calibrating *Cyano Vision* to actual cyanobacterial vision became methodologically valuable in itself; each failed parameter adjustment exposed how deeply my sensory assumptions were embedded in human visual norms.



☹️ The apparatus functioned as an unlearning tool, revealing how persistently human-centred my imagination remained. Moving forward, I recognised that *Cyano Vision*, rather than serving solely as an exploratory artefact, operated as a speculative probe. Its value lay in its exposure of anthropocentric assumptions and in unsettling habitual modes of perceiving microbial worlds. By foregrounding the limitations and frictions in simulating cyanobacterial sensing, *Cyano Vision* functioned as a device for unlearning; it prompted reconsideration of what it meant to encounter difference and invited ongoing reflection on human interpretative frames. The technical challenge was also an ethical one: how to design for becoming-with rather than becoming-as.





These parallel trajectories—one pursuing controlled motility, the other attempting perceptual translation—converged on three interrelated insights about practising biodesign with MTH sensibilities.

Wearing the prototype of *Cyano Vision* forced me to confront what I had assumed as an individual territory was a singular shared space. The laboratory exists at radically different scales; what appears to me as benchtop and incubator constitutes an entirely different spatial reality for microbes inhabiting gradients I cannot perceive with my naked eye. This scalar difference, present in everyday laboratory encounters, shaped how I navigated my relationship with the environment and its inhabitants. Each parameter adjustment made the familiar environment increasingly alien, revealing how profoundly my understanding privileges human-scale perception. **Designing with microbial awareness requires learning to hold these incommensurable perspectives together, not collapsing their alterity into human frameworks, but developing approaches that acknowledge differences of scale as constitutive of shared environments themselves.**

The augmented setup made visible the entangled agencies constituting even the simplest protocol. **I had imagined myself as the sole actor orchestrating movement, but repeated failures revealed a dense multispecies ecology:** my body maintaining temperature and humidity through breath and presence; the organisms, each an autonomous agent; contaminating fungi and bacteria claiming inadequately sterilised plates; and non-living instruments mediating our encounter. **The protocol was never a human directing microbes, but a contingent assemblage of bodies, technologies, and environmental conditions.** The relational texture of these encounters became apparent through situated practice, as every attempt to shape experimental conditions demanded attention to the web of influences at play. Recognising this entanglement transformed my practice from instrumental control toward response-ability: learning to attune to the micro-scale requirements that make cyanobacterial agency possible, understanding that biodesign always involves negotiating among multiple, often conflicting, forms of livingness.

Both trajectories exposed the laboratory as operating across temporalities radically divergent from human and microbial lifeworlds. My struggle to observe motility unfolded against their circadian rhythms, evolutionary lineage, and the minutes-to-hours scale of their phototactic response. Each failed experiment represented weeks on my timeline but encompassed entire lifecycles of contaminating organisms and subtle physiological shifts in the cyanobacteria I sought to understand. The lab ecology operates simultaneously across multiple temporal registers, from photosynthetic processes occurring in microseconds to culture growth measured across days, strain maintenance spanning decades, and evolutionary processes extending across deep time. These overlapping temporalities revealed the laboratory as a space where rhythms and durations intermingle, demanding attentiveness to how biodesign practice is shaped by encounters with time at many scales. Observing beyond the laboratory revealed them as agents reshaping planetary chemistry across geological scales. This temporal diversity demands practices that acknowledge temporal differences as fundamental to MTH engagement.

Different perspectives on the laboratory

Single cell, 70 trillion cells, and others

Laboratory as a unique ecosystem

So, how can biodesigners integrate MTH sensibilities into their practice?

What did I learn along the way?

The two autoethnographic accounts conducted in a biodesign laboratory demonstrate that cultivating MTH sensibilities within environments typically governed by sterility, regulation, and meticulous microbial handling is not only possible but can unfold along multiple trajectories. MTH learning can emerge not only through protocol-driven laboratory practice but also through experimental and speculative inquiry in hybrid studio–laboratory settings.

Before the divergence into two trajectories, my approach was grounded in a fundamental understanding of organisms that resonated with traditional biological training. For designers lacking a background in biology, particularly those accustomed to working with inert materials such as plastic, metal, or wood, the bottom-up biodesign approach that starts from a scientific understanding of organisms can initially feel unfamiliar, even coercive. Living materials, however, resist predictability; they demand technical precision, ethical attentiveness, and continuous responsiveness. This scientific rigour, while preserving biological integrity, also creates the conditions for a distinctive form of MTH learning that is unique to biodesign.

Scientific knowledge plays a crucial role in supporting MTH sensibilities in biodesign. Biodesign necessarily involves working with living systems across theoretical and applied regenerative contexts, so biological and ecological knowledge helps designers move beyond anthropocentric assumptions. It provides both a technical foundation and a cultivated respect for livingness at the microbial scale.

In biological experiments, neglecting the precise environmental requirements for cyanobacterial phototactic motility—such as temperature, humidity, and agar concentration—revealed how easily anthropocentric assumptions fill epistemic gaps. In studio-lab explorations, defining the optical qualities of cyanobacteria primarily through human visual perception led prematurely into speculative design. In both cases, scientific knowledge provided a corrective scaffold for MTH learning, helping prevent misdirection and enabling more responsible design possibilities.

However, blind assimilation into scientific workflows carries its own risks. As demonstrated in the laboratory journey, the bio-digital setup initially developed to understand phototaxis gradually became instrumentalised for control and application. In laboratories prioritising sterility and scientific equivalence to conventional biology labs, their borrowed instrumental practices can dominate the environment, subtly compelling biodesigners to conform to utilitarian and technocentric logics.

While scientific tools illuminate mechanisms, they rarely address the broader entangled relationships in which organisms are embedded. For instance, although biodesigners may claim to “collaborate” with microbes, any “uninvited” microorganism is quickly labelled as “contamination” and excluded from the process. When unknown fungal or bacterial colonies appear on plates, they are typically discarded immediately, without consideration of their possible relevance or relationship. Echoing Anna L. Tsing’s observation that “As contamination changes world-making projects, mutual worlds—and new directions—may emerge” (Tsing 2015, 27), this prompts a crucial question: Do such selective monoculture biodesign practices truly support regenerative ecologies and nurture MTH sensibilities? While cultivating *Staphylococcus aureus*, Alexander Fleming (1929) observed a contaminating fungus inhibiting bacterial growth. Recognising its potential significance, he preserved and studied it, ultimately leading to the discovery of penicillin. The pressure to translate living qualities into socially applicable artefacts further reinforces anthropocentric design paradigms and restricts opportunities to enact MTH-oriented values, for example, by encouraging attunement to microbial conditions rather than technical optimisation.

To unlearn this intrinsic instrumentalisation, the laboratory journey suggested the value of shaping more exploratory and self-reflective design tools for biodesigners. Within the strict confines of laboratory environments, overhauling established scientific infrastructures may be unrealistic. However, by starting with familiar laboratory artefacts and questioning their taken-for-granted assumptions, I found that even subtle reconfigurations could be significant, for instance, by adapting existing protocols to facilitate closer observation of microbial behaviour or by creating opportunities for ethical reflection within everyday experimental routines. Instead of focusing on spectacular MTH applications, such as high-tech installations or colourful displays, these efforts open opportunities for situated reflection and the expression of MTH thinking within real constraints.

In contrast, when design inquiry begins with exploratory imagination outside the conventional laboratory workflow, MTH approaches can be more actively foregrounded. Hybrid environments that blend laboratory work with external studio spaces enable engagement with speculative and HCI-informed methods. Immersive digital tools, such as Meta Quest 3, supported the development of *Cyano Vision*, facilitating embodied and sensory experiences that narrow the perceptual distance from microbial entities. For organisms whose cognitive or sensory attunement is otherwise inaccessible, extended or virtual sensory mediation can serve as a bridge, much like scientific knowledge itself.

Yet another tension emerges here. As shown, speculative freedom may inadvertently marginalise scientific rigour, which can be detrimental when working with living systems. Drawing from the challenges I encountered in prototyping, biodesigners’ need for mechanisms to continually assess whether their design processes accurately reflect the biological and ecological qualities of the organisms involved became noticeable. These mechanisms do not suppress ethical awareness; rather, they counteract implicit anthropocentrism and encourage critical unlearning.

Along the way, the process of unlearning began to manifest in subtle but tangible shifts within my design practice. Rather than approaching laboratory work as a matter of imposing control, I found myself increasingly attuned to unfolding conditions and emergent dynamics as they arose. Encounters with contamination, once met with the instinct to eradicate, became occasions to reconsider boundaries and interactions; still attentive to safety, but open to the possibility of learning through these interruptions. In developing sensory simulations, my focus gradually shifted from purely replicating microbial phenomena to translating across sensory modalities, allowing new forms of understanding to emerge. These changes, while not always dramatic, signal how unlearning can quietly recalibrate the designer’s everyday actions and sensibilities.

Looking back: What became visible through my self-reflective (un)learning?

Throughout the preceding accounts, the narration has moved between immediacy and reflection, not simply to chronicle events but to foreground how personal encounters with uncertainty, adaptation, and failure become sources of insight. Rather than offering a purely descriptive record, the autoethnographic voice is used to draw out the underlying patterns and shifts in orientation that emerged from these experiences; patterns that speak not just to my own process, but to broader dynamics at play in biodesign practice. This mode of narration thus serves to distil, from the lived detail, those recurring tensions and inflexion points that shape the trajectory of MTH engagement.

Adopting an explicitly subjective voice enabled the surfacing of relational dynamics that conventional laboratory documentation tends to conceal. Previously, my descriptions of cyanobacteria were confined to strain codes, italicised taxonomies, and binary states: alive, dead, dormant, viable. Such documentation, shaped by a controlled laboratory culture, left little room to articulate MTH reflections.

Through autoethnographic narration, I was able to articulate the network of relations among microbes, tools, protocols, and myself. Cyanobacteria emerged not merely as research subjects but as living organisms encountered in situated interaction. This reframing unsettled habitual modernist separations between object and subject and opened space for unlearning entrenched laboratory hierarchies.

Importantly, the value of this approach lies not in presenting a universal MTH tool but in offering a nuanced account of tensions, frictions, and negotiations. Because biodesign contexts vary widely—in organisms, purposes, infrastructures, and institutional cultures—such positioned narratives provide relatable entry points for other biodesigners seeking to apply MTH thinking within their own constraints.

Did my journey happen in solitude?

While this inquiry is narrated from my perspective and adopts an autoethnographic voice, its realisation was fundamentally collective. Over the past three years, my team—spanning microbiology, HCI, MTHD, and (bio)design—has consistently engaged in reflective sessions on my journey that we referred to as “retreats.” These gatherings became forums for sharing diverse disciplinary perspectives, critically examining methodologies, and collaboratively navigating the evolving process. They not only surfaced and negotiated interdisciplinary tensions within the field but also provided the scaffolding needed to move beyond methodological boundaries. Without such sustained, dialogic reflection, I might have remained confined within my own assumptions, unable to fully engage with or appreciate the scientific and MTH dimensions of biodesign.

Epilogue and Impact statement for biodesign

This study demonstrates that MTH sensibilities in biodesign are not confined to speculative or activist settings but can also emerge within scientifically rigorous laboratory environments. Instead of positioning scientific precision and relational awareness as oppositional, this autoethnographic account illustrates how their dynamic interplay fosters productive reflection and innovation. By tracing moments of instrumentalisation, misalignment, and recalibration, the study illuminates the often-unspoken negotiations and tensions that shape everyday biodesign practice, offering a richer understanding of how MTH perspectives can be integrated into the discipline.

Methodologically, this work extends biodesign research by employing autoethnography as a critical tool for self-reflection and analysis. Articulating the first author’s relationship with cyanobacteria beyond conventional scientific labelling reveals the underlying power dynamics and epistemic hierarchies present in laboratory culture. This reflective approach does not dismiss scientific rigour; rather, it positions science and technology as active collaborators in cultivating MTH sensibilities. By broadening the methodological vocabulary of biodesign, this perspective suggests that meaningful transformation can arise not only from new materials or artefacts but also from reconfiguring the ways practitioners narrate, interpret, and relate to their engagements with living systems.

Practically, this study provides insight into how existing laboratory tools and workflows can be subtly reoriented to support working with cyanobacteria without compromising safety or scientific precision. It encourages interdisciplinary collaboration among technicians, microbiologists, and laboratory managers to collectively reshape the conditions under which biodesign is practised. By fostering this collaborative reorientation, the study expands the methodological boundaries of the field and strengthens biodesign’s capacity to engage regenerative futures with greater vigilance, responsibility, and adaptability.

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Competing Interests

The authors have no conflicts of interest to declare for this publication.

Data Availability Statement

The primary experimental materials generated and used in this study comprise (1) an adapted macroscopic phototaxis assay of *Synechocystis* sp. PCC 6803, and (2) a Unity application package for *Cyano Vision*.

(1) Data associated with the adapted phototaxis assay is available at <https://doi.org/10.4121/d3466d24-9012-4b35-8818-e4535cda60bd>.

(2) The Unity application package for *Cyano Vision* is available at <https://github.com/OEOxr/CyanoVision-MR>.

Please note: Both items were developed for exploratory research and have not undergone formal validation. For further information or access to experimental materials, contact the corresponding author.

Ethical Standards

Ethical approval and consent are not relevant to this article type.

Author Contributions

Conceptualisation: Jiho Kim, Elvin Karana. Data curation: Jiho Kim. Formal analysis: Jiho Kim. Investigation: Jiho Kim. Methodology: Jiho Kim, Elvin Karana. Project administration: Elvin Karana, Joana Martins. Resources: Elvin Karana. Software: Okan Efe Ögretmen. Supervision: Elvin Karana, Joana Martins, Iohanna Nicenboim. Visualisation: Jiho Kim. Writing original draft: Jiho Kim. Writing review & editing: All authors.

Note

I. This illustration was produced for commentary articles accompanying a press release on the eLife research article “Cyanobacteria use micro-optics to sense light direction” (Schuergers *et al.* 2016). It does not appear in the original research article and is only available in the associated commentaries.

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